Documentation and operational readiness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask what another operator can verify, who owns failure and which recovery is compatible.

Teach complete operational instructions with secret-free records and honest rehearsal status.

Keep one habit: Model setup gaps → follow healthy/unavailable cards → inspect missing owner → assess recovery dependencies → record performed versus unperformed checks.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of documentation and operational readiness

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of red team exercises

Reference example, not the only acceptable wording: Teacher explanation: Red-team exercises probe explicit failure boundaries in an authorised isolated fixture environment. Their purpose is **evidence**-based improvement.

Use fictional records and allowed test routes. Define a target contract violation, record the exact case, observe action/data traces and preserve a reproducible report. Do not touch external systems.

Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of performance and cost experiments

Reference example, not the only acceptable wording: Teacher explanation: Performance experiments compare matched workloads with declared timing boundaries and quality constraints. Averages can hide slow cases.

Use the same cases and environment; record every duration and success. For a simple median sort values and average the middle pair for even counts. Keep failed requests visible.

Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of incident investigation

Reference example, not the only acceptable wording: Teacher explanation: Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues.

Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate containment from diagnosis. Compare candidate explanations against **evidence**.

Pictured end goal

Explain documentation and operational readiness using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Red team exercises

Performance and cost experiments

Incident investigation

Use the three-session teaching cycle above, with extra time as needed. Operational documentation tells another person how to run, verify, observe and recover the service. A successful developer demo is not readiness. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of documentation and operational readiness

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach reproducible setup, bounded operation and honest rehearsal evidence

> A developer demo and a polished guide do not establish every readiness condition.

Definitions and complete fictional runbook

Operational documentation tells another person how to run, verify, observe and recover the service. A successful developer demo is not readiness.

A **runbook** names versions, required non-secret configuration, secret key names, **health check**s, release/rollback dependencies, alerts and owners. Test it with a fresh operator and fictional fixtures.

Runbook element	Complete fictional rehearsal reference
Reviewed combination	Service code toy-r2, config cfg 2, corpus c1, procedure p2, schema 1, fixture F1
Non-secret config	ENV=classroom-mock; DATA_VERSION=c1; read-only source card F1
Secret names, never values	SESSION_KEY is a name-only documentation example; paper mock requires no secret
Preparation	Copy supplied F1 card below; record versions; make no external connection
Healthy F1 card	Approved room B12; corpus c1; source_available=true

Runbook element	Complete fictional rehearsal reference
Verification	Check versions and card availability; expected supported fixture answer B12 with source F1/c1
Unavailable variant	source_available=false; expected SOURCE_UNAVAILABLE and needs_review, no invented room
Owner and route	Symbolic maintainer role; prepare incident packet for approved internal route, not an actual message
Fallback	Keep unsupported request unresolved; do not broaden credentials or use an unapproved source
Recovery dependencies	Reviewed toy-r1 reads schema 1/c1 only; check current schema/version before a proposed rollback
Rehearsal status	Not run unless an actual partner follows the instructions; paper self-trace is labelled separately

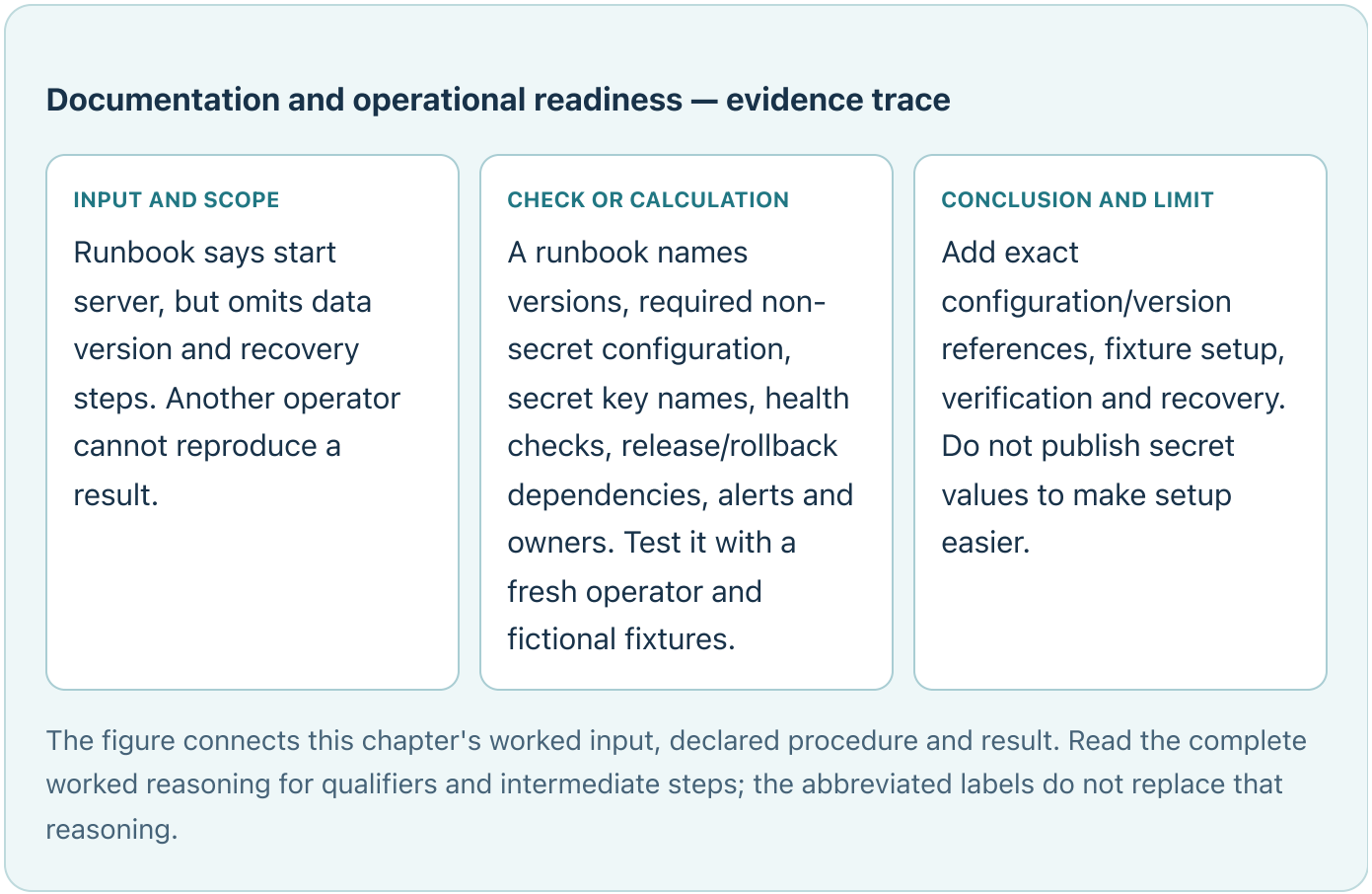
Main gap ledger and checked reasoning

Missing from the main setup-only note	Why another operator needs it	Concrete supplied repair
Code/config/data versions	Otherwise two operators may use different evidence	toy-r2 / cfg 2 / c1 / p2 / schema 1 / F1
Fixture and expected result	Cannot distinguish start-up from supported service behavior	F1 approved B12; record source F1/c1
Unavailable-source check	Cannot know honest failure behavior	SOURCE_UNAVAILABLE → needs_review, no unsupported room
Owner and permitted route	Readable diagnostics alone leave unresolved responsibility	Named symbolic maintainer and prepared incident packet
Compatible recovery conditions	Old code may not read current data	Check toy-r1 supports current schema 1/c1; schema 2 would block
Actual rehearsal record	A developer demo is not another operator’s successful use	Record performed/not run, first confusion, evidence and revision

Main “start server” note lacks versions, fixture verification and recovery. Added complete fiction records toy-r2/cfg 2/c1/p2/schema 1/F1, healthy approved B12 and unavailable SOURCE_UNAVAILABLE → needs_review. Document secret names only; paper exercise needs no secret values. Owner/route and compatibility checks are part of operation, not optional polish.

Picture removing maintainer leaves responsibility missing despite readable diagnostics. Toy-r1 supports schema 1/c1 only; schema 2 variant blocks. No actual diagnostic, contact, deployment or rollback is performed. Only record a fresh-operator rehearsal if a real partner follows the steps; self-trace is different.

Selected setup-only case needs incident owner/fallback and tested compatible rollback dependencies, not another setup paragraph.



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

Qualification: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Runbook says start server, but omits data version and recovery steps. Another operator cannot reproduce a result.

Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized operational **evidence** without secret values or unnecessary private content so a later revision can be compared honestly.

Version, secret and rehearsal-status probes

Readiness probe	Expected interpretation
Service starts but data version absent	Reproducibility incomplete
Secret values pasted into guide	Remove values; document names and approved acquisition process separately
Healthy card gives B12 with F1/c1 evidence	Passes that fictional check only
Unavailable source yields guessed room	Fails supported-answer fallback
Diagnostic readable but owner absent	Escalation responsibility missing
Old recovery reader cannot read current schema	Block incompatible proposal; use separately tested compatible plan
Developer says it worked	Not evidence another operator completed the runbook
Partner unavailable	Record self-trace or not run; do not invent rehearsal sign-off
Runbook rehearsed on F1	Scoped evidence, not readiness for every incident

Separate unavailable-source symptom and owner

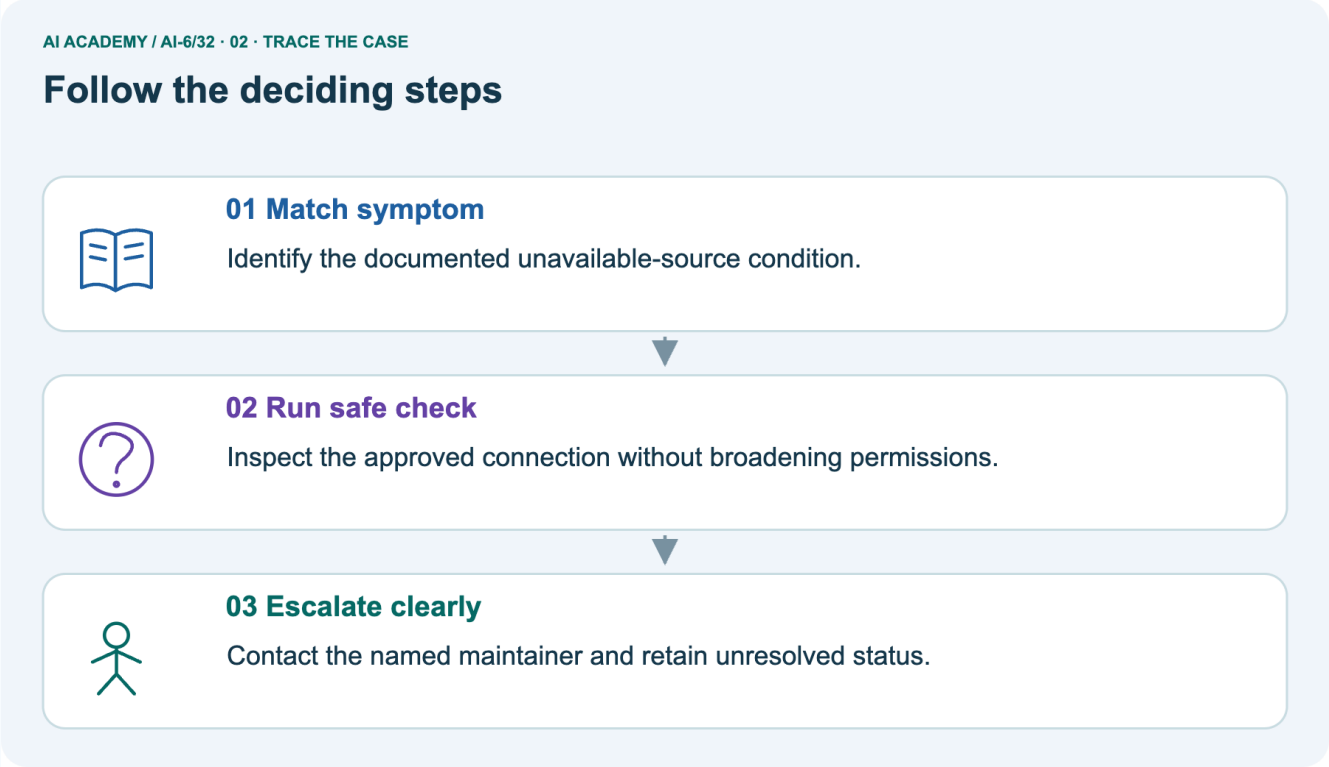
A **runbook** with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer.

The **runbook** names a maintainer and a read-only source-connection diagnostic. The code means no supported answer can be produced.

The picture focuses on SOURCE_UNAVAILABLE. Under this fictional contract the unavailable source means no supported answer can be produced; keep needs_review rather than guess. The documented read-only check inspects only the approved source connection/configuration, without granting broader permissions.

The named maintainer and authorized route give the unresolved incident an owner; actual contact must be separately performed and recorded. Removing them leaves readable diagnostic steps but no complete responsibility/escalation path. It does not authorize the operator to invent a new owner, secret or answer. A later recovery still needs its own verification against the recorded versions.

Picture trace and missing-route key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Match symptom	Identify the documented unavailable-source condition.
2. Run safe check	Inspect the approved connection without broadening permissions.
3. Escalate clearly	Contact the named maintainer and retain unresolved status.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Operational documentation connects a symptom, bounded action, responsible person and recovery check. Component names alone do not tell an operator what to do.

Rehearse and revise **runbook**s; readable instructions alone cannot establish readiness for every incident.

Change: Remove the maintainer and escalation route.

Prediction: The diagnostic remains, but responsibility for an unresolved fault becomes unclear.

Reason: Readiness includes who acts when routine checks do not solve the problem.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Runbook	Instructions for operating and recovering a service
Health check	A defined observation of service state
Operational owner	The actor responsible for ongoing decisions

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

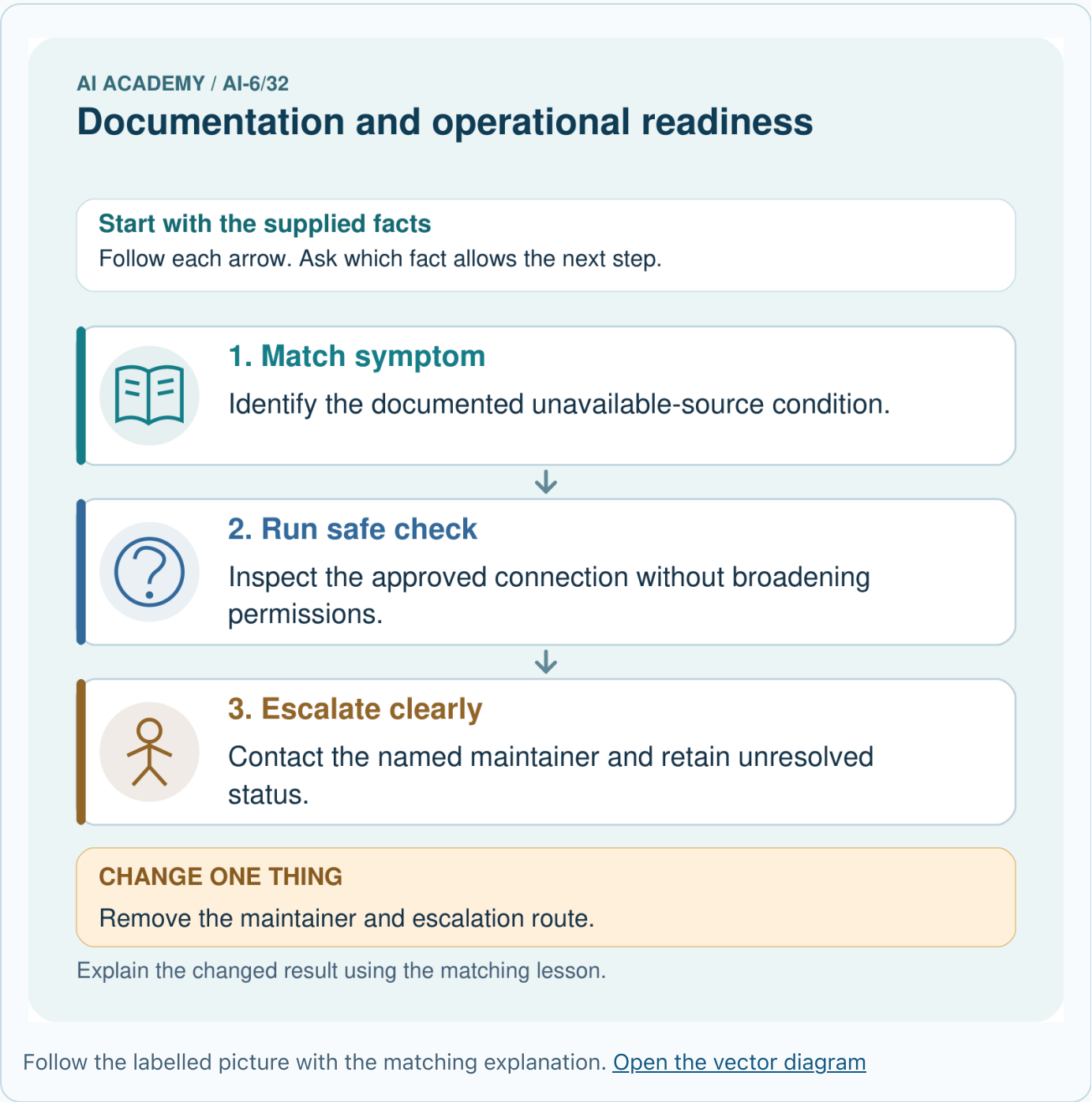
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Documentation and operational readiness

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: **Runbook** says start server, but omits data version and recovery steps. Another operator cannot reproduce a result. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Documentation and operational readiness. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Match SOURCE_UNAVAILABLE to its safe check and escalation owner. What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?

Identify the documented unavailable-source condition. Inspect the approved connection without broadening permissions. Contact the named maintainer and retain unresolved status.

Changed case: The diagnostic remains, but responsibility for an unresolved fault becomes unclear. Readiness includes who acts when routine checks do not solve the problem.

Changed-case reasoning: The diagnostic remains, but responsibility for an unresolved fault becomes unclear. Readiness includes who acts when routine checks do not solve the problem.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main runbook needs exact versions, fixture setup, expected verification and compatible recovery. Added paper F1 rehearsal expects B12 with source evidence when available and needs_review/SOURCE_UNAVAILABLE when not; no actual operations or messages. Picture missing maintainer route leaves responsibility unresolved. Selected setup-only guide still needs incident owner, fallback and tested rollback dependencies.

Record actual partner rehearsal, self-trace and unperformed checks distinctly; never invent sign-off or publish secret values.

Diagnose the current misconceptions

The worked result establishes every future case: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Documentation includes all setup steps but no incident owner or compatible rollback. Identify readiness gaps.

Teacher explanation: Specify escalation owner, fallback and tested rollback dependencies. A setup-only guide does not cover operation.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Operational documentation tells another person how to run, verify, observe and recover the service. A successful developer demo is not readiness.

A **runbook** names versions, required non-secret configuration, secret key names, **health check**s, release/rollback dependencies, alerts and owners. Test it with a fresh operator and fictional fixtures.

W2 Trace the worked case

Runbook says start server, but omits data version and recovery steps. Another operator cannot reproduce a result. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

W3 Audit the limit

A report omits this qualification: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

Qualification: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Documentation includes all setup steps but no incident owner or compatible rollback. Identify readiness gaps.

Teacher explanation: Specify escalation owner, fallback and tested rollback dependencies. A setup-only guide does not cover operation.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Specify escalation owner, fallback and tested rollback dependencies. A setup-only guide does not cover operation. Require intermediate reasoning and the stated scope.

Fresh retry: **Runbook** contains an old corpus ID after release. Expected reasoning: Update version-bound instructions and independently replay them; preserve earlier release documentation.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled documentation and operational readiness **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of documentation and operational readiness.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Capstone service implementation. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/33, ai-6/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of documentation and operational readiness

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: **Runbook** says start server, but omits data version and recovery steps. Another operator cannot reproduce a result. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

Check every learner before discussion: Use the worked case for Documentation and operational readiness. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Documentation readiness requires a tested replay and **operational owner**ship, not merely complete-looking prose.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Remove the maintainer and escalation route. Require a **prediction** and a reason before comparing. Expected result: The diagnostic remains, but responsibility for an unresolved fault becomes unclear.

Reason: Readiness includes who acts when routine checks do not solve the problem.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

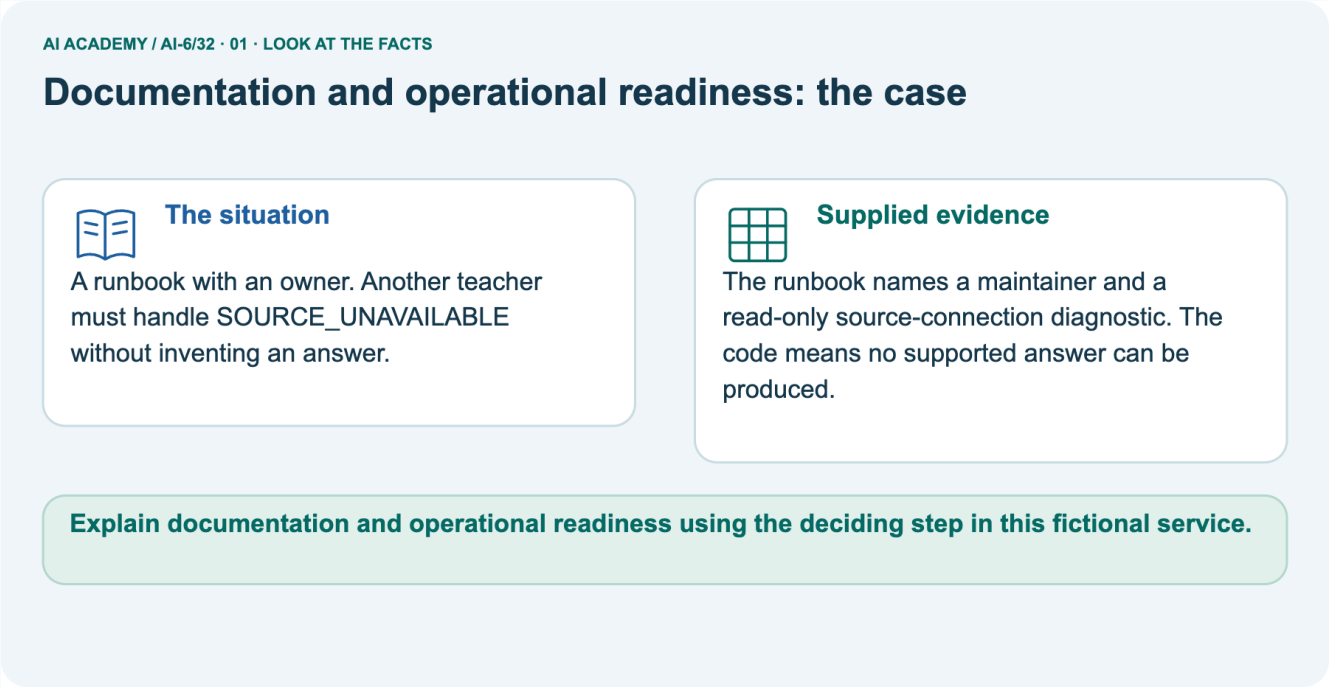
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Documentation and operational readiness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

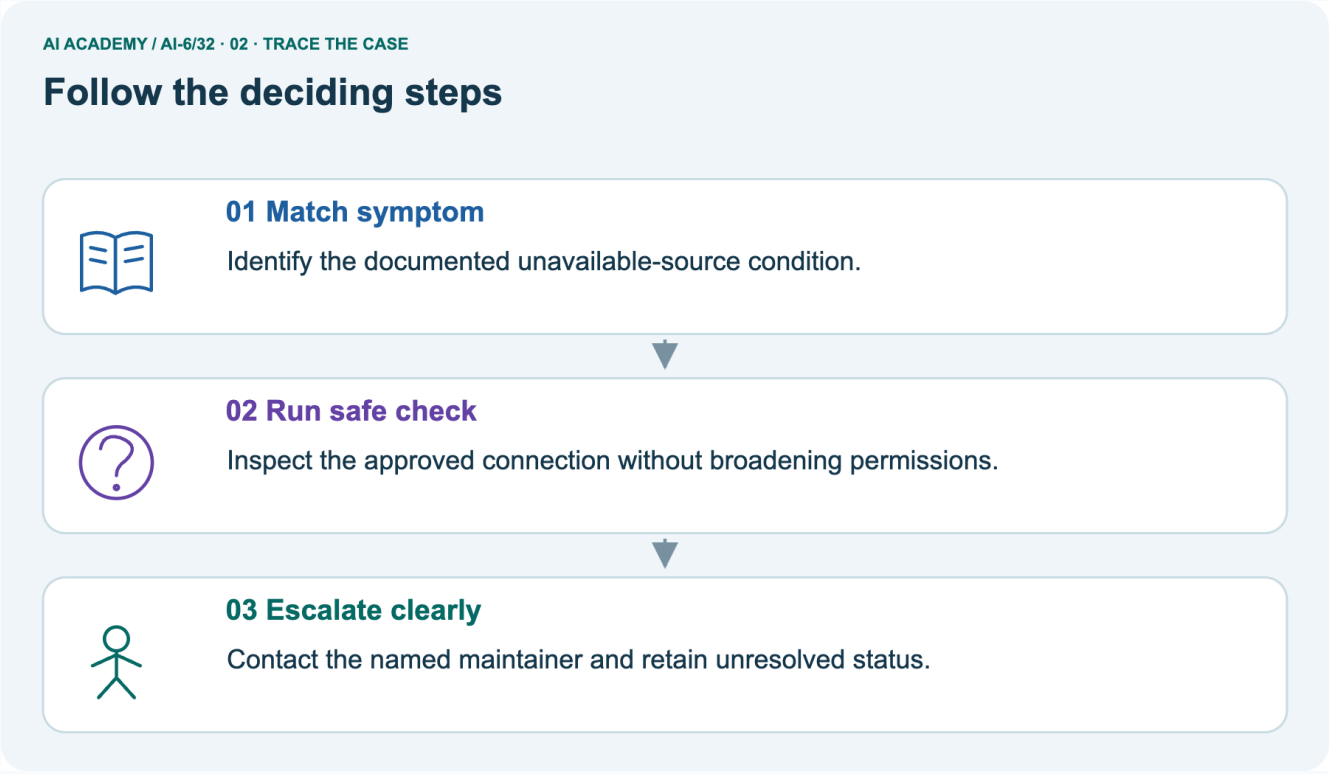
A **runbook** with an owner. Another teacher must handle SOURCE_UNAVAILABLE without inventing an answer. Supplied facts: The **runbook** names a maintainer and a read-only source-connection diagnostic.

The code means no supported answer can be produced.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



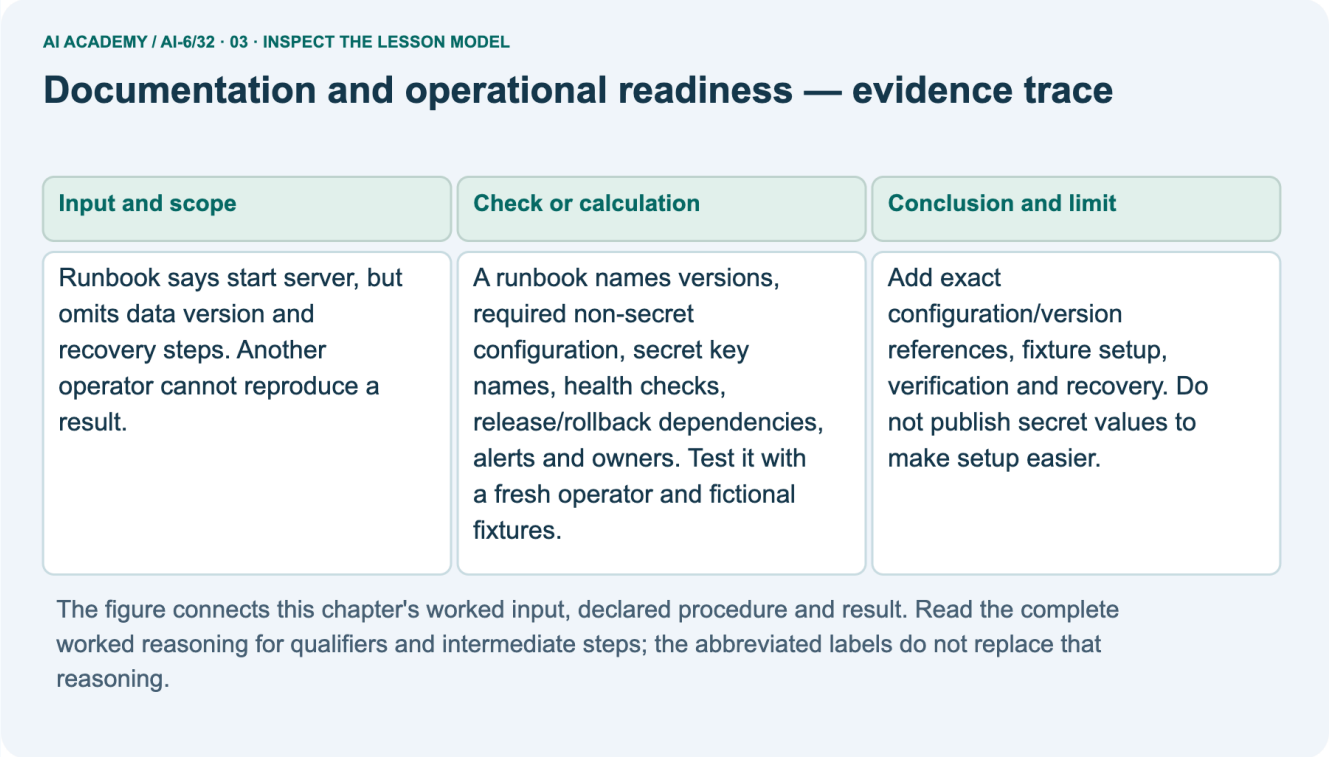
Follow the labelled picture with the matching explanation. Open the vector diagram

Match symptom: Identify the documented unavailable-source condition. → Run safe check: Inspect the approved connection without broadening permissions. → Escalate clearly: Contact the named maintainer and retain unresolved status.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Documentation and operational readiness — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Documentation and operational readiness — **evidence** trace. **Runbook** says start server, but omits data version and recovery steps.

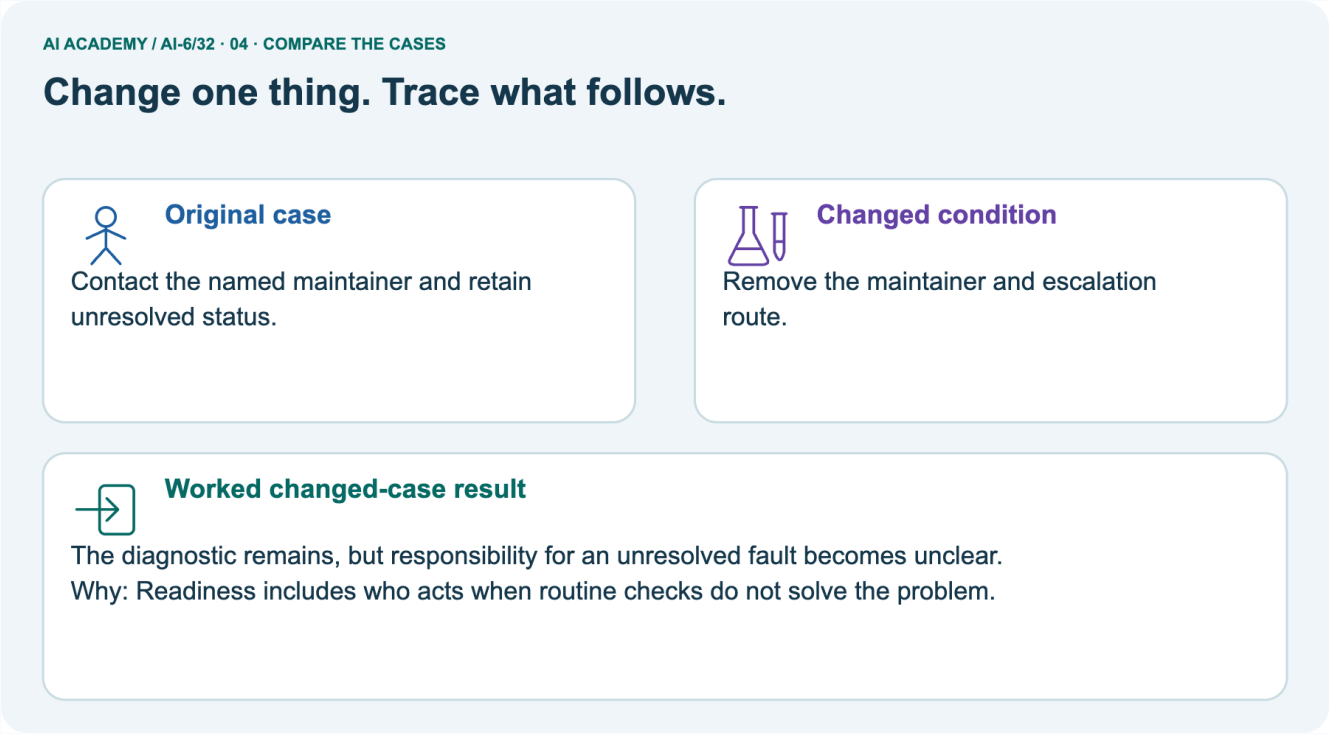
Another operator cannot reproduce a result. | A **runbook** names versions, required non-secret configuration, secret key names, **health check**s, release/rollback dependencies, alerts and owners.

Test it with a fresh operator and fictional fixtures. | Add exact configuration/version references, fixture setup, verification and recovery. Do not publish secret values to make setup easier.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Contact the named maintainer and retain unresolved status. Changed condition: Remove the maintainer and escalation route.

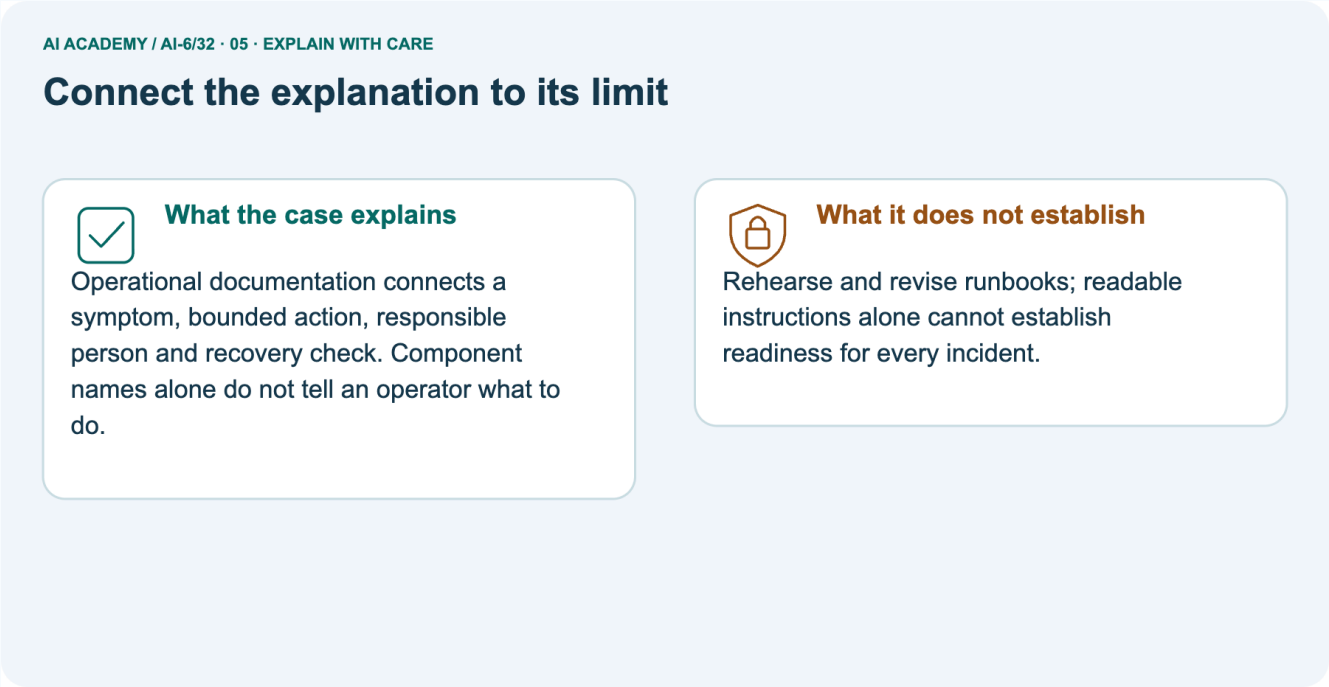
Expected result: The diagnostic remains, but responsibility for an unresolved fault becomes unclear. Reason: Readiness includes who acts when routine checks do not solve the problem.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Readiness includes who acts when routine checks do not solve the problem.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Operational documentation connects a symptom, bounded action, responsible person and recovery check. Component names alone do not tell an operator what to do.

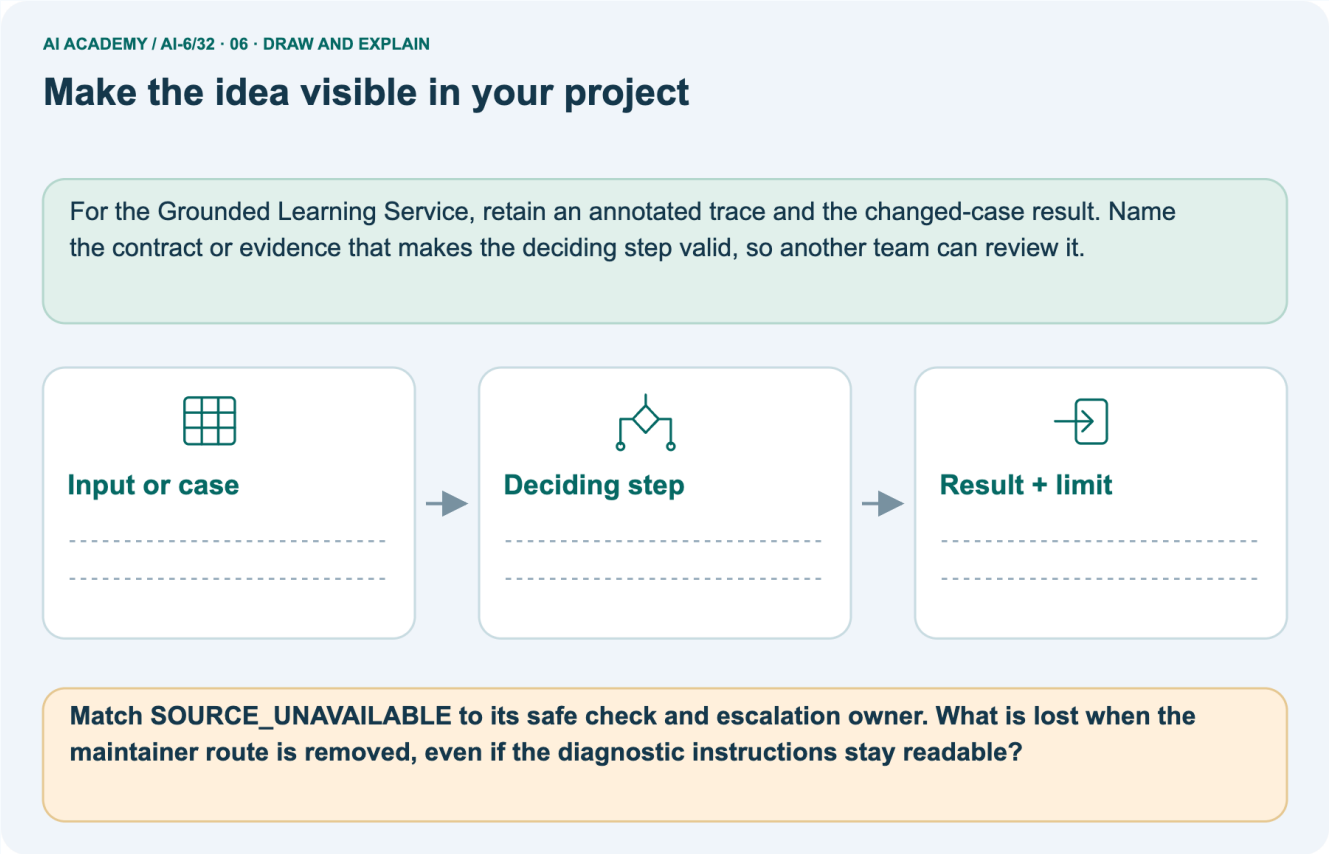
Limit: Rehearse and revise **runbook**s; readable instructions alone cannot establish readiness for every incident.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Rehearse and revise **runbook**s; readable instructions alone cannot establish readiness for every incident.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Identify the documented unavailable-source condition. Inspect the approved connection without broadening permissions. Contact the named maintainer and retain unresolved status.

Changed case: The diagnostic remains, but responsibility for an unresolved fault becomes unclear. Readiness includes who acts when routine checks do not solve the problem.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Documentation and operational readiness: the case

02. Follow the deciding steps

03. Documentation and operational readiness — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.